

ASSIGNMENT GUIDELINE

Rules

- This is a group assignment, minimum of **3** and maximum of **4** students.
- The marks allocated for this assignment is **20%** of your total coursework.
- Due date to submit report: **15 May 2026 (Fri, Week 7)**

The objectives of this assignment are to:

- Encourage students to lead and participate in teamwork with good communication skills.
- Develop students' practical understanding of virtualization technology.
- Analyse and benchmark the performance of virtual machines within the VirtualBox environment.

Content Guideline

1. Virtual Machine Setup

Each group must create at least **three** or **four virtual machines** with different configurations (e.g., different OS, RAM, CPU cores, storage allocation, etc.). Example configurations:

- **VM1:** Ubuntu with 2 CPU cores, 4GB RAM, 20GB storage
- **VM2:** Fedora with 1 CPU core, 2GB RAM, 30GB storage
- **VM3:** Ubuntu with 2 CPU cores, 8GB RAM, 20GB storage
- **VM4:** FreeBSD with 2 CPU core, 4GB RAM, 40GB storage

The group members may discuss and define their own configurations.

2. Performance Metrics for Comparison

Each group must evaluate and compare the following performance metrics:

- **CPU Usage:** Measure and compare CPU utilization under different workloads.
- **Memory Usage:** Observe RAM consumption for each VM.
- **Disk Performance:** Compare read/write speeds using benchmarking tools.
- **Boot Time:** Measure the time taken for each VM to boot up.
- **Application Performance:** Run a simple application (e.g., web server, web browser, code editor) and record performance differences.

3. Performance Testing Tools

Students should use tools such as:

- **VirtualBox built-in performance graphs**
- **Task Manager (Windows) / System Monitor (Linux)**
- **htop (Linux)**
- **CrystalDiskMark (Windows) / GNOME Disks (Linux)**
- **Other tools which are suitable, e.g. Sysbench**

Report Guideline (20%)

- 1) Cover page
- 2) Table of contents
- 3) Body of report.
 - i. Introduction: Brief explanation of virtualization and the importance of VM performance analysis.
 - ii. VM Setup Details: Table comparing VM configurations.
 - iii. Methodology: How tests were conducted and what tools were used. Explain what applications are installed in each VM for comparison purpose.
 - iv. Results & Analysis: Graphs, tables, and explanations comparing the performance of the VMs.
 - v. Conclusion: Key findings and recommendations for optimal VM performance.
 - vi. Appendix: Supporting documents/ Articles
 - vii. Format Report:
 - Font: Times New Roman, 12 pt.
 - Spacing: 1.5
 - Justify
 - Header:
 - Top Left corner: TCC4223 Introduction to Cloud Computing
 - Top Right corner: Assignment
 - Footer:
 - Bottom Left corner: Tri March/April 2026
 - Bottom Right corner: Page number
- 4) Please use the cover page in the last page of this document.
- 5) Minimum pages of the report are 8-10 pages.
- 6) Originality of content is the most important within this assignment. You can use any similarity checker application to check.
- 7) Submission will be in softcopy with PDF format as Assignment_Group Number_TCC4223_2610 (i.e. Assignment_Group1_TCC4223_2610.pdf).



FACULTY OF INFORMATION SCIENCE AND TECHNOLOGY

**TCC4223
INTRODUCTION OF CLOUD COMPUTING**

ASSIGNMENT

Trimester March/April 2026 (Term 2610)

No.	Student ID	Student Name	Set VM	Signature
1.				
2.				
3.				
4.				